

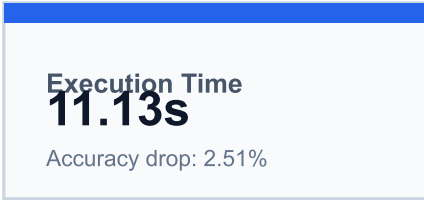
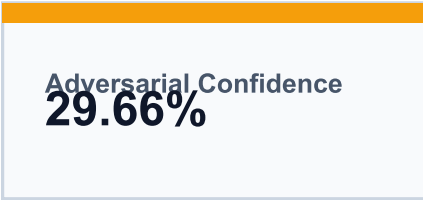
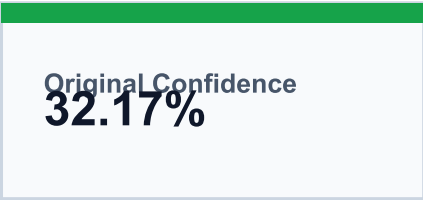
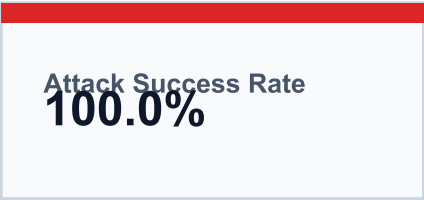
Threat Assessment Report

Executive summary for model robustness under adversarial attack

Model: microsoft/resnet-18

Attack Type: FGSM

Images Evaluated: 10



Threat Level Assessment

HIGH RISK

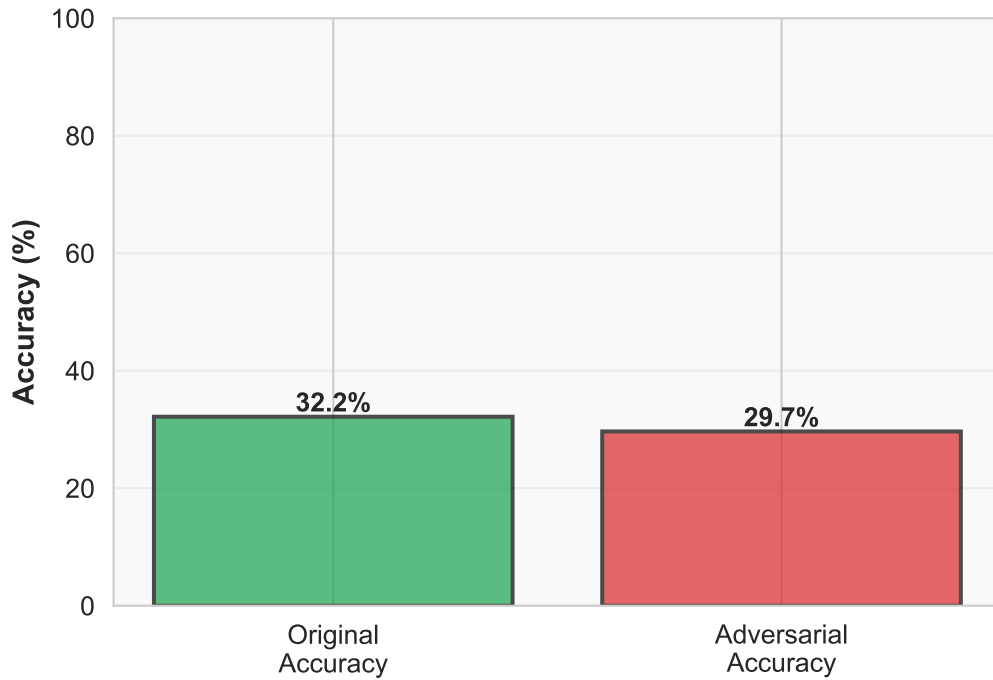
The FGSM attack altered model predictions on 100.0% of the evaluated images. Average confidence shifted from 32.17% to 29.66%, indicating the current robustness posture is high risk.

Assessment Narrative

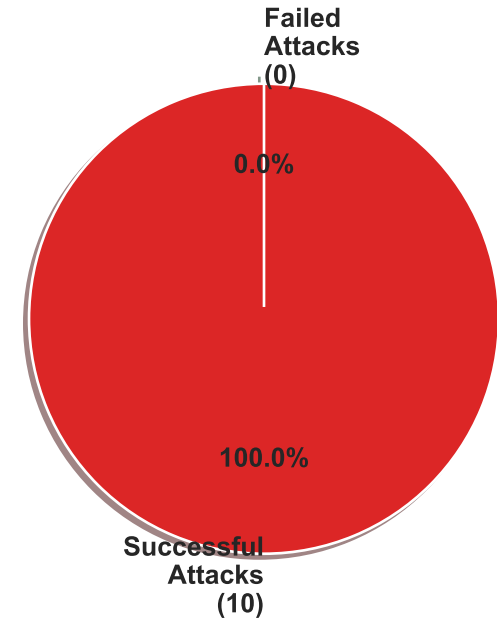
Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 100.0%. Average original accuracy: 32.17%, Average adversarial accuracy: 29.66%. The attack successfully fooled the model in 10 out of 10 cases.

Detailed Analysis

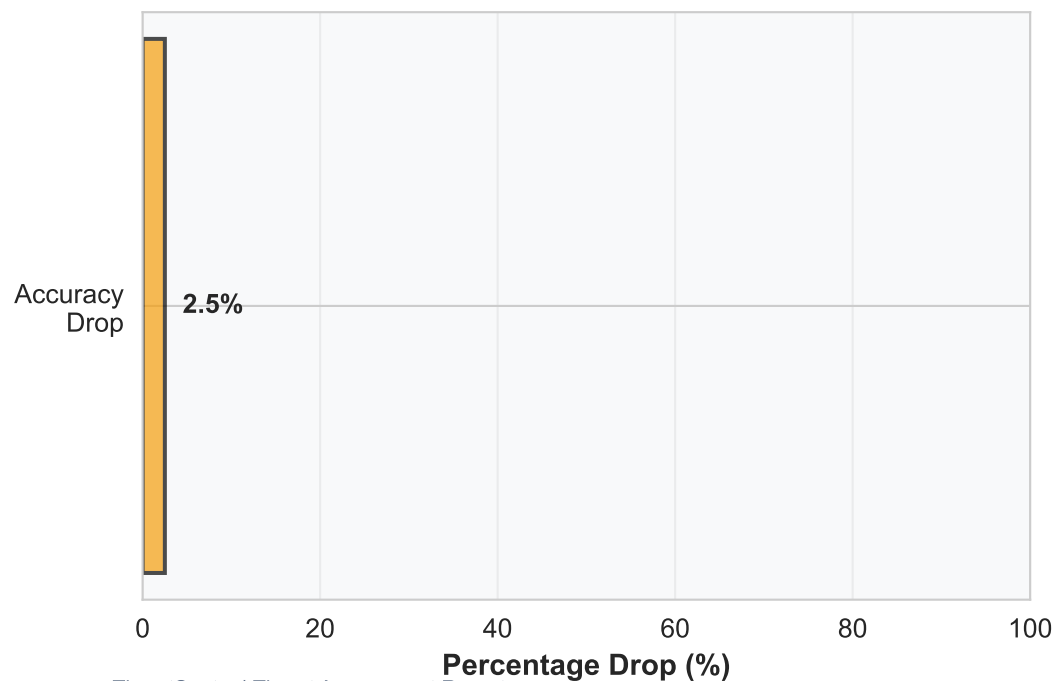
Accuracy Comparison



Attack Success Distribution



Model Robustness Impact



Confidence Comparison (Sample)

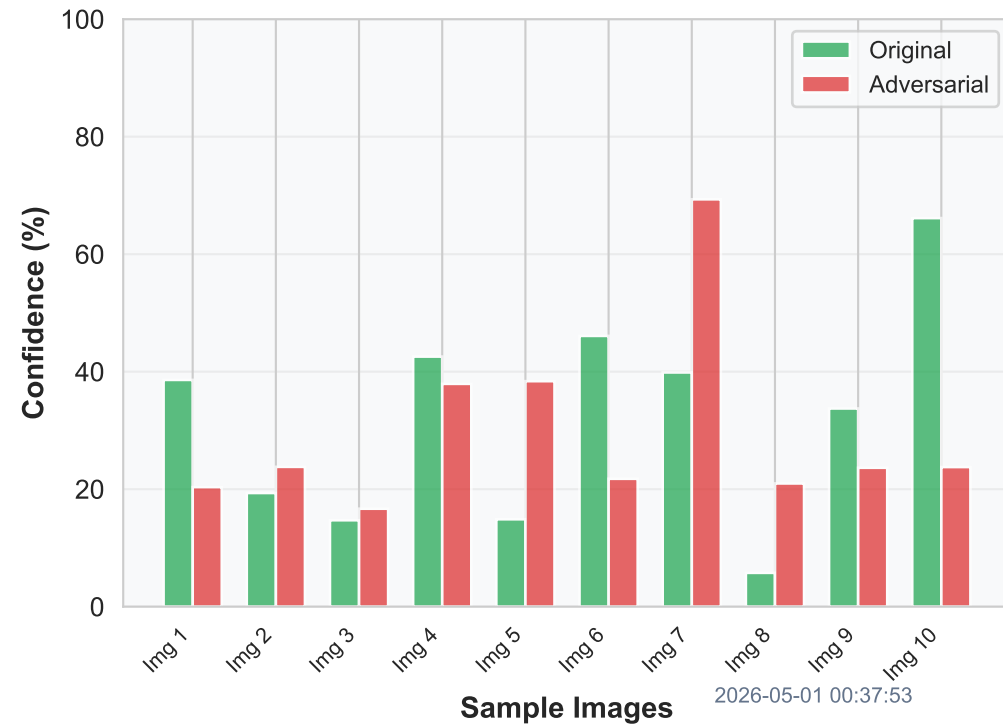


Image Evidence and Predictions

Images 1-4 of 10

Image preview unavailable

Attack Successful

Original image: 24264.jpg
Adversarial image: 24264.jpg
(attacked)
Outcome: Attack Successful
Model predicted as: alp
Original confidence: 39.87%
After attack predicted as: jigsaw
puzzle
Adversarial confidence: 69.34%

Image preview unavailable

Attack Successful

Original image: 24252.jpg
Adversarial image: 24252.jpg
(attacked)
Outcome: Attack Successful
Model predicted as: alp
Original confidence: 46.09%
After attack predicted as: volcano
Adversarial confidence: 21.77%

Image preview unavailable

Attack Successful

Original image: 24251.jpg
Adversarial image: 24251.jpg
(attacked)
Outcome: Attack Successful
Model predicted as: maze, labyrinth
Original confidence: 5.76%
After attack predicted as: American
alligator, Alligator mississippiensis
Adversarial confidence: 20.98%

Image preview unavailable

Attack Successful

Original image: 24256.jpg
Adversarial image: 24256.jpg
(attacked)
Outcome: Attack Successful
Model predicted as: mountain bike,
all-terrain bike, off-roader
Original confidence: 14.71%
After attack predicted as: shower
curtain
Adversarial confidence: 16.67%

Image Evidence and Predictions

Images 5-8 of 10

Image preview unavailable

Attack Successful

Original image: 24261.jpg
Adversarial image: 24261.jpg
(attacked)
Outcome: Attack Successful
Model predicted as: ice bear, polar bear, Ursus Maritimus, Thalarctos maritimus
Original confidence: 42.57%
After attack predicted as: American alligator, Alligator mississippiensis
Adversarial confidence: 27.63%

Image preview unavailable

Attack Successful

Original image: 24262.jpg
Adversarial image: 24262.jpg
(attacked)
Outcome: Attack Successful
Model predicted as: castle
Original confidence: 19.35%
After attack predicted as: bell cote, bell cot
Adversarial confidence: 23.80%

Image preview unavailable

Attack Successful

Original image: 24258.jpg
Adversarial image: 24258.jpg
(attacked)
Outcome: Attack Successful
Model predicted as: palace
Original confidence: 33.74%
After attack predicted as: drilling platform, offshore rig
Adversarial confidence: 23.65%

Image preview unavailable

Attack Successful

Original image: 24260.jpg
Adversarial image: 24260.jpg
(attacked)
Outcome: Attack Successful
Model predicted as: worm fence, snake fence, snake-rail fence, Virginia fence
Original confidence: 14.88%
After attack predicted as: American alligator, Alligator mississippiensis
Adversarial confidence: 38.39%

Image Evidence and Predictions

Images 9-10 of 10

Image preview unavailable

Attack Successful

Original image: 24263.jpg
Adversarial image: 24263.jpg (attacked)
Outcome: Attack Successful
Model predicted as: valley, vale
Original confidence: 66.14%
After attack predicted as: alp
Adversarial confidence: 23.77%

Image preview unavailable

Attack Successful

Original image: 24254.jpg
Adversarial image: 24254.jpg (attacked)
Outcome: Attack Successful
Model predicted as: swing
Original confidence: 38.60%
After attack predicted as: shower curtain
Adversarial confidence: 20.35%

Recommendations

Security actions based on observed threat exposure

Assessment Summary

Successfully executed FGSM attack on model microsoft/resnet-18 using 10 test images. Attack success rate: 100.0%. Average original accuracy: 32.17%, Average adversarial accuracy: 29.66%. The attack successfully fooled the model in 10 out of 10 cases.

Security Recommendations

1. Implement Adversarial Training

- Retrain your model with adversarial examples to improve robustness
- Use techniques like FGSM, PGD during training phase

2. Add Input Validation & Preprocessing

- Implement input sanitization and anomaly detection
- Use defensive distillation or feature squeezing

3. Deploy Ensemble Methods

- Use multiple models with different architectures
- Implement voting mechanisms for predictions

4. Continuous Monitoring

- Set up real-time performance monitoring
- Detect and alert on unusual prediction patterns

5. Regular Security Audits

- Conduct periodic threat assessments
- Stay updated with latest attack techniques